

Detection, Tracking, and Interdiction for Amateur Drones

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ABSTRACT

Unmanned aerial vehicles, also known as drones, are expected to play major roles in future smart cities, for example, by delivering goods and merchandise, serving as mobile hotspots for broadband wireless access, and maintaining surveillance and security. The goal of this survey article is to study various potential cyber and physical threats that may arise from the use of UAVs, and subsequently review various ways to detect, track, and interdict malicious drones. In particular, we review techniques that rely on ambient radio frequency signals (emitted from UAVs), radars, acoustic sensors, and computer vision techniques for detection of malicious UAVs. We present some early experimental and simulation results on range estimation of UAVs and receding horizon tracking of UAVs. Finally, we summarize common techniques that are considered for interdiction of UAVs.

INTRODUCTION

Enabled by recent technological advances, miniaturization, and open source hardware/software initiatives, use of unmanned aerial vehicles (UAVs) for a wide range of applications is quickly transforming from a futuristic idea to reality. Pervasive use of drones for future smart cities, however, also raises several technical and societal concerns and challenges that must be addressed, as UAVs can be used to conduct physical attacks (e.g., via explosives) and cyber attacks (e.g., hacking a critical infrastructure). Vital information can be obtained from drones by using network exploits and malware-based attacks. Drones with high dependency on GPS can be hijacked using GPS spoofing, to be used against people and infrastructure [1].

As the cost of commercial small UAVs is decreasing, the number of hobbyists using them is increasing rapidly. Thus, it is becoming difficult to identify unauthorized drones and bring them down when necessary. Small drones are a danger to civilian aircraft, and there have been reports of near accidents involving UAVs and aircraft around airports. It is critical that small UAVs are restricted from entering protected areas, and an unauthorized UAV that may be a potential threat should be detected, interdicted, and brought down in a safe area.

In this article, we provide a unified review of the techniques for detecting, tracking, and interdicting small unauthorized UAVs around restricted areas. As shown in Fig. 1, we consider that UAV detectors are deployed around the area of

interest; the detectors can be ground units as well as other monitoring UAVs (see, e.g., the *Aerial Dragnet* concept from the Defense Advanced Research Projects Agency, DARPA, in the United States). If the trajectory of the UAV remains outside the protected area, it is considered to be safe. If the trajectory crosses into the protected area, the drone needs to be detected and brought down outside the protected zone.

Our main contribution in this article is to provide a unified overview of different techniques to detect and track UAVs: RF signals radiated from UAVs, radar, acoustic signals, and computer vision methods. We also summarize the main physical and cyber threats from unauthorized drones, as well as radio propagation channel considerations related to UAV detection/tracking. Preliminary experimental results are provided considering ultra-wideband radar-based range estimation/tracking to a small drone. We present a *receding horizon* approach for tracking of malicious UAVs by other UAVs, and conclude the article with an overview of common drone interdiction approaches.

THREATS FROM UNAUTHORIZED DRONES

In this section we review some common cyber and physical threats introduced by unauthorized drones, giving examples from popularly used small commercial UAVs.

PHYSICAL THREATS FROM UNAUTHORIZED DRONES

A major threat from small UAVs is their use in direct physical attacks on people, property, and infrastructure. Small UAVs may, for example, carry explosives, radioactive materials, or guns that may be used by terrorist groups, countries, and even individuals with malicious intent, to carry out targeted attacks on people and infrastructure. For instance, in 2015 a quadcopter crashed on the lawn of the White House, which (despite being unintentional) shows the potential dangers that can be introduced by unauthorized small UAVs. Organized crime groups can use small UAVs to carry illegal information or material, such as smuggling drugs across national borders. Targeted physical attacks to a country's critical infrastructure, such as its power grid, can enable follow-up cyber attacks.

Another major concern related to the use of drones is that of privacy. It is easy to mount a camera or recording device to capture information, and this can violate personal privacy. To overcome such concerns, the Center of Democracy and Technology (CDT) has asked the FAA

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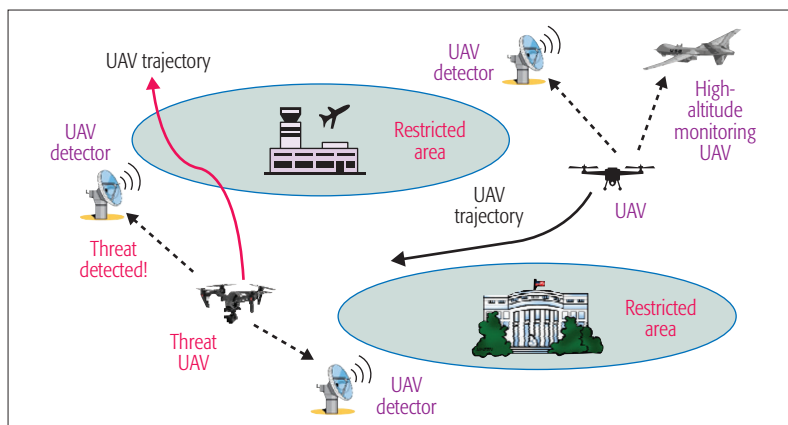


Figure 1. Detection of unauthorized small UAVs around restricted areas.

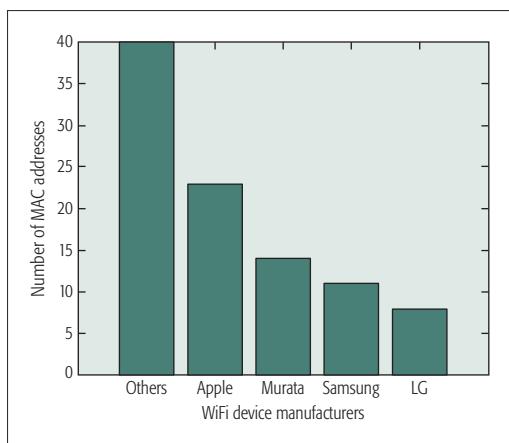


Figure 2. Unique devices identified from different manufacturers based on the WiFi probe requests captured at the UAV.

to issue rules on privacy, and recommends using a data collection statement to know whether the information collected will be retained, used, or disclosed. The most suitable means to maintain privacy was termed Privacy by Design (PbD), which helps in maintaining standards and provides remedies for security breaches. By adopting PbD principles, privacy intrusion becomes limited, and privacy can be ensured at an early stage [2].

CYBER THREATS TO/FROM UNAUTHORIZED DRONES

Due to their agile nature, drones can be used to approach the vicinity of a cyber-physical system to perform cyber attacks, such as hijacking (e.g., other drones), spoofing, “sniffing,” and hacking. For example, many commercial, lower-end drones use WiFi technology for communication between the drone and ground station controller, which may be vulnerable to security breaches and attacks from other drones. An unencrypted Wi-Fi link used with lower-end commercial drones such as the Parrot AR.Drone allows any individual to connect and potentially hack the drone (see e.g., *Skyjack* software) [2]. Moreover, a de-authentication attack can be carried out by one drone to other drones, e.g., using the *aircrack-ng* software which sends disassociate packets to connected clients (such as those remotely controlling the drone) for disconnecting them from the drone’s Wi-Fi controller connection. The popular DJI Phantom 3 is also susceptible to such Wi-Fi attacks, while (to the best knowledge of

the authors) such weaknesses have not yet been reported in the newer DJI Phantom 4 drone.

Another way drones are commonly used is to sniff (eavesdrop) wireless signals that may include private information. For example, a small UAV such as a DJI Phantom, with a WiFi Pineapple mounted on it, can be used to easily sniff wireless signals from peoples’ houses. Built-in tools such as *airdroid-ng*, *sslstrip* can be used to dissect the wireless data. The basic idea is to land the drone on a building/balcony to collect data, and afterward bring it back to the originating location. The connection with the WiFi Pineapple can be maintained over a 3G connection by creating an SSH relay server.

The data captured by the WiFi Pineapple is time stamped and includes MAC address, received signal strength indicator (RSSI), and manufacturer’s name of the WiFi enabled devices. Such data can be used to identify and track individuals in public space (e.g., based on daily/weekly behavior patterns), which can introduce privacy and security concerns. Based on the WiFi probe requests captured, Fig. 2 provides an example for the number of different types of devices that we have detected around a UAV’s flight trajectory around Florida International University. In just a few minutes, over one hundred unique mobile WiFi devices were identified. Such data collected at city scale and longer time durations by unauthorized drones may have serious privacy implications.

DETECTION OF UNAUTHORIZED DRONES

In this section, since a good majority of drone detection techniques rely on the use of RF signals, we first discuss some key RF propagation characteristics that may impact drone detection/tracking accuracy. We then review key techniques for drone detection: ambient RF signals, radar, acoustic signals, computer vision, and data fusion. Advantages/disadvantages of these different techniques are summarized in Table 1.

RADIO PROPAGATION CHANNEL CONSIDERATIONS

As is well known, radio propagation characteristics are frequency-dependent. Thus, knowledge of the specific frequency band(s) used by drones for signaling is a prerequisite for effective detection. The primary channel characteristic of interest for detection and jamming is attenuation, because for a given value of transmit power and receiver sensitivity, path loss determines the detection or effective jamming range. Since path loss also depends on the environment, path loss models, and consequently detection range, are also environment-specific. For example, path loss differs dramatically between open rural settings and dense (or “cluttered”) urban environments [3]. Often, path loss models specify the presence or absence of a line of sight (LOS) between transmitter and receiver, so one should account for the LOS or non-LOS (NLOS) condition in evaluating detection or jamming. For drones, the presence/absence of LOS is a direct function of elevation angle and communication link type (e.g., air-ground or air-air). Traditional air-ground channels typically have relied on LOS, and most modern applications of drones also aim to place the aircraft well above ground, hence increasing the probability of LOS. In such cases, diffraction can be more prevalent than shadowing, although for very low altitudes, terrestrial obstacle

Detection Technique	Advantages	Disadvantages
Ambient RF signals (e.g., [4, 6])	Low-cost RF sensors (e.g., SDRs), works in NLOS, long detection range. May allow deauthentication attacks for taking control of drone by mimicking a remote controller or spoofing GPS signal.	Need prior training to identify/classify different drones. Fails for fully/partially autonomous drone flights due to no/limited signal radiation from a drone/controller.
Radar (e.g., [7–10])	Low-cost FMCW radars, does not get affected from fog/clouds/dust as opposed to vision based techniques, can work in NLOS (more sophisticated). Higher (mmWave) frequencies allow capturing micro-Doppler/range accurately at the cost of higher path loss. Does not require active transmission from the drone.	Small RCS of drone makes identification/classification difficult. Further research needed for accurate drone detection/classification and machine learning techniques, considering different radar/drone geometries and different drone types which all affect micro-Doppler signatures. Higher path loss at mmWave bands limits drone detection range.
Acoustic signals (e.g., [11–13])	Low cost for simple microphones (cost depends on the quality of microphones). Can work in NLOS as long as the drone is audible.	Need to develop database of acoustic signature for different drones. Knowledge of current wind conditions and background noise is needed. May operate poorly under high ambient noise such as in urban environments.
Computer vision (e.g., [9, 14, 15])	Low cost for basic optical sensors. Pervasive availability of cameras even at most commercial drones that can be used as sensors.	Higher cost for thermal, laser-based, and wide FOV cameras. Requires LOS. Level of visibility impacted by fog, clouds, dust.
Sensor fusion (e.g., [9, 15])	Can combine advantages of multiple different techniques for wider application scenario, high detection accuracy, and long-distance operation.	Higher cost and processing complexity. Need effective sensor fusion algorithms.

Table 1. Comparison of advantages/disadvantages for different drone detection and tracking techniques.

shadowing models may be used. Additional work is required for specific drone shadowing models in various settings.

Multiple path loss models exist for the unlicensed bands in different environments. Most of these are for cellular and indoor applications, but to first order, for low-altitude (and slow-moving) drones, the cellular models can be used as initial approximations. But new air-ground path loss models for low-altitude drones, in multiple environments, should be developed. The obvious reason for this is that drones will not only hover in high locations away from local obstacles (mimicking cellular base stations), but will also move in three dimensions throughout an environment. Hence, effects such as diffraction and waveguiding may be more important for some drone operational conditions than in typical cellular channels. For LOS channels with no clutter very near the ground receiver, quasi-deterministic “two-ray” models have been shown to be fairly accurate [3], with measured standard deviations from the log-distance linear fit less than 3 dB. These models have an LOS component and Earth surface reflection, which can be estimated from geometry, along with a number of multipath components that are modeled statistically.

Also of interest is channel dispersion attributable to multipath propagation. For example, in some cluttered settings, a strong multipath reflection may be the only detectable signal from a low-power drone at low altitude. This raises the topic of antenna directionality. The use of directional antennas can increase link range and suppress, or enhance, multipath components. This potential advantage comes with the potential disadvantage of a narrower field of view, requiring more time if the drone signal must be found by searching spatially. As higher frequency bands are used, more sophisticated arrays for fast multiple-beam scanning will become feasible.

The mobility of drones also has effects on the propagation channel. Mobility causes spatial variation of signal intensity as well as Doppler effects. Since Doppler shifts are directly proportional to both velocity and carrier frequency, as these increase, Doppler shifts can be significant, particularly for narrowband signals. As an example, for a carrier frequency at the

low end of the millimeter-wave (mmWave) band (e.g., 30 GHz), a small drone moving at its top speed of, say, 30 m/s, can cause a Doppler shift of 3 kHz. This may be an appreciable fraction of a narrowband signal's bandwidth (and is approximately 25 percent of the current LTE subcarrier bandwidth).

RADIO FREQUENCY SIGNALS FROM UAVS

In [4], possible solutions and challenges of inexpensive RF-based drone detection were discussed, and experimental low-cost feasibility demonstrations were provided. Detection methods were classified as:

1. Analyzing the reflected signal from the propellers
2. Eavesdropping on the communication between the drone and its controller
3. Analyzing the reflection patterns from vibration of the drone's body.

It was shown via experiments that the first approach is feasible for distances below 3 m, and the second approach works for distances below 50 m. These limits, of course, depend on established transmit power limits and the propagation environment. In [5], it was shown that the hopping sequence in popular frequency hopping spread spectrum physical layer signals can be extracted using software defined radios (SDRs). Since most commercial radio controlled UAVs use this type of physical layer, getting access to the hopping sequence opens the way to not only detect and track these UAVs more reliably, but also to jam and spoof them. Unique RF transmission patterns from popular commercially available UAVs can be used to train a classifier (e.g., a neural network), which can then be used to classify a specific UAV in real time.

In [6], three prevalent commercial RF physical layers for remote UAV control are characterized. An analog FM transmitter's radio footprint and communication protocol were compared with two more advanced methods. The spread spectrum implementation was decoded and shown to have an RF footprint distinguishable from that of FM. These two were further compared with a DJI Phantom 3 link, and it was suggested that bandwidth and central frequency can be used to detect and classify the type of UAV. Although most commercially available small drones employ unlicensed fre-

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quency bands and often WiFi signaling, with the prevalence and continued growth of SDRs, knowledgeable drone users will eventually configure their own radio links, with customized physical layers, and if needed, customized medium access control (MAC) layers. Depending on signaling complexity, this could be possible in nearly any frequency band from very high frequency (VHF) bands up to (at present) approximately 6 GHz. Such novel radio signaling will present much more challenging cases for detection.

RADAR

When there is no active RF radiation from a drone (e.g., fully/partially autonomous operation), or if such RF signals cannot be detected reliably, use of RF based radar techniques can be an appealing alternative for detecting/tracking drones. As opposed to, say, optical sensors, radars are active sensors, and they can operate day or night. The mmWave radars, in particular, have very large bandwidths, and hence have a high resolution that can facilitate accurate detection, classification, and tracking of unauthorized UAVs. A major challenge with radar-based detection of *micro-drones* is that they have very small radar cross section (RCS), and they fly at lower altitudes and at lower speeds compared to larger drones [7]. Use of mmWave radars enables one to precisely analyze the Doppler spectrum, which can provide information for classifying the type of the drone, understanding whether they carry any payload (e.g., explosives), and distinguishing them from other flying objects/animals such as birds.

Use of radar techniques for detection, classification, and tracking of micro-drones has been recently explored in the literature. For example, in [7], the authors considered the use of multi-static radars operating at 2.4 GHz to analyze micro-Doppler signatures of micro-drones (DJI Phantom Vision 2) with different payloads, and classify drones into no payload, 200 g payload, and 500 g payload categories. A transmit power around 23 dBm was used, along with horizontally polarized antennas with 24 dBi gain and $10^\circ \times 10^\circ$ beamwidth, at radar to target distances less than 100 m. The proposed feature detection techniques consistently achieve more than 90 percent classification accuracy.

In [8], frequency-modulated continuous-wave (FMCW) radar operating at 35 GHz was used to classify DJI Phantom 2 (quadcopter) and DJI S1000 (octocopter) drones, at radar ranges of 30–90 m. Authors observed wide Doppler side-lobes in range-Doppler profiles due to the high carrier frequency. On the other hand, information about the drone's range, radial velocity, size, type, and even shape can be extracted (also altitude if two antennas are used). The authors in [9] presented an mmWave radar operating at 94 GHz at 20 dBm transmit power, a bandwidth of 1 GHz, and a range resolution of 15 cm. The radar can cover distances from 10 m up to several hundred meters. In [10], the authors considered the use of a 16-receive, 16-transmit multiple-input multiple-output (MIMO) radar to generate 256 virtual elements to detect and track drones. The radar operates at 36 GHz central frequency, and uses pulses of duration 500 ns. A hypothesis testing framework was developed to decide whether a drone exists within a certain range/angular cell. Experimental results

by the authors show that the target drone can be tracked accurately at distances exceeding 100 m and at radial velocity of about 5 m/s.

For proof of concept purposes, we conducted preliminary experiments with Time Domain P410 UWB radar equipment to evaluate range estimation accuracy to a small quadcopter. The UWB transceivers have an instantaneous bandwidth of 2.2 GHz (between 3.1 GHz and 5.3 GHz), transmitted power from the radio was limited to -14.5 dBm to comply with FCC regulations, and at the receiver a sampling resolution of 61 ps is utilized. A large number of channel impulse response (CIR) realizations based on the reflected radar signals from the UAV were captured at the UWB radar receiver considering a static channel environment, and consecutive CIRs were subtracted from each other to filter out the multipath components from static objects. From these data, statistics (e.g., variance and mean) of the Doppler shifts can be calculated, which are then used to find likelihood functions and subsequently range estimates at each time instant using a maximum likelihood estimator; these estimates are illustrated with white dots in Fig. 3a, which are observed to be discontinuous across different timescales. Finally, a Kalman filter was used to track the range of the target UAV as shown by the red curve in Fig. 3b. Results show that with time domain Kalman filtering, the noisy range estimates can be smoothed to improve accuracy. Range estimates at multiple radar locations can be used to triangulate a target UAV's location. Directional antennas and beamforming techniques can be used to increase the radar detection range.

ACOUSTIC SENSORS

Acoustic processing methods are a seemingly obvious method of detecting and tracking small UAVs. After all, we are all too familiar with the distinctive buzzing noise made by a small UAV that is flying nearby. However, there is more than meets the ear to these acoustic methods. Several research groups [11–13] have considered acoustic sensors and have demonstrated the ability to detect small UAVs (quad-copters) in ranges from 20 m [11] up to 600 m [12] with a 99.5 percent probability of detection and a 3 percent false positive rate.

The two main differentiators in prior work are the signal processing method and the type of acoustic sensor employed. In terms of sensors, microphone arrays [11–13] have some advantages over a single microphone: they offer not only the chance to detect the presence of a drone, but through beamforming they can also find its azimuth and elevation, thus allowing for angle of arrival localization by using two or more microphone arrays. Additionally, the coherent detection enabled by beamforming also increases the signal-to-noise ratio, which in turn increases the range and rate of correct detection and tracking. The authors of [12] used a tetrahedral configuration for the microphones with each microphone at the tips of a tetrahedral pyramid with microphones approximately 27 cm apart from each other, showing a 99.5 percent probability of detection for ranges smaller than 600 m. In contrast, in [13], a commercial product (Distran Omni360) used 120 elements in a spherical arrangement to reliably detect drones between 150 m and 290 m depending on the drone type.

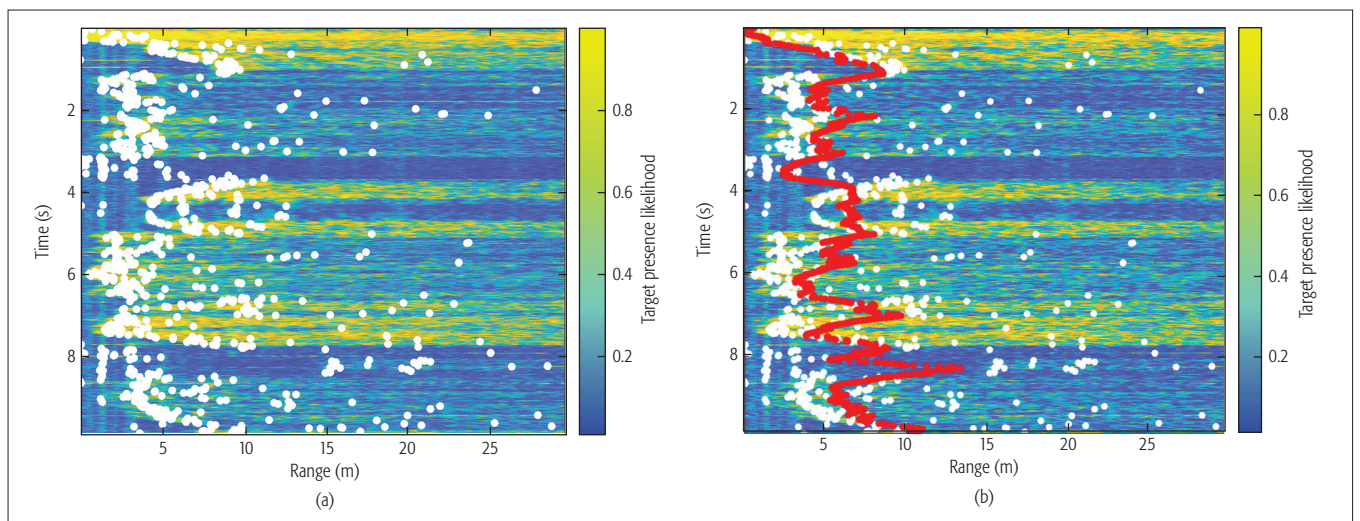


Figure 3. Range estimation to a UAV using UWB radar: a) distance likelihood function; b) distance likelihood function with time domain filtering.

Signal processing methods range from searching for a known signature [11–13] using the well-known signal processing technique of spectrograms, to deep neural networks trained specifically to distinguish small UAVs from background noise. In terms of advantages, acoustic detection and tracking can be completely passive, and relatively inexpensive (although system performance does benefit up to a point from an increase in the number and quality of microphones as well as the overall computational power). Range of reliable detection depends not only on the type of acoustic sensors, but also on environmental factors, for example, the presence and intensity of wind as well as background noise. Additionally, accurate detection and tracking relies on a comprehensive database of acoustic signatures, which in turn is vulnerable to new methods of propulsion with signatures that may differ substantially from the ones in the database.

COMPUTER VISION

Another approach for detecting and tracking unauthorized drones is computer vision. In the last few years, a wide variety of long-standing problems in machine learning, artificial intelligence, and computer vision have seen significant improvements. Average detection precision at the ImageNet Large Scale Visual Recognition Competition (ILSVRC) had reached 0.80 in 2016 from 0.22 in 2013. Detection and tracking of drones based on computer vision has important advantages, including:

1. Being able to detect drones that do not have RF transmission
2. Use of passive and cheap optical sensors
3. Excellent inherent directional accuracy

For faster and more accurate detection, visual and thermal cameras with different fields of views (FOVs) can be used simultaneously: wider FOV can be used for coverage, while narrower FOV can be used for better detection/classification accuracy. Infrared and laser-based techniques are alternative optical approaches for detection of drones. For example, for detecting drones at night, use of shortwave infrared cameras was recently demonstrated in [14].

A major disadvantage of optical sensors is that they operate poorly when visibility is limited (e.g., in the presence of clouds, fog, and dust). They

also cannot operate in NLOS conditions such as in urban environments. To improve the detection accuracy in such scenarios, camera-based techniques are typically used in combination with other earlier discussed detection methods, such as acoustic sensors [15] and radar [9].

SENSOR FUSION

As should be apparent from earlier discussions, different technologies available for detection and tracking of UAVs have various trade-offs related to cost, accuracy, precision, range, energy efficiency (critical if sensors operate on batteries), portability (e.g., sensors deployed at other UAVs), and complexity (Table 1). For example, while some sensor technologies can only operate very well in LOS conditions (e.g., computer vision), some others can also operate in NLOS environments (e.g., RF-based approaches). For accurate and quick detection/tracking of UAVs, data fusion techniques that can simultaneously use information from multiple types of sensors carry critical importance (see, e.g., [9, 15] for joint use of 94 GHz mmWave radar and acoustic sensors, respectively, with high resolution optical cameras), and this constitutes an open research area.

TRACKING OF UNAUTHORIZED DRONES

Using signal strength observations from an unauthorized target drone, it is possible to localize and track them. To this end, fixed sensors on the ground, as well as one or more other UAVs can be used (Fig. 1). In [16], we consider the use of multiple UAVs to localize and track a mobile UAV, with the goal of achieving the optimal Cramer-Rao lower bound on the localization error through maximization of the *D-optimality criterion*. In particular, we use a Fisher information matrix (FIM) predictive model and a receding horizon maximization technique to plan feasible paths for the tracking UAVs that lead to the best achievable localization error bound. In other words, the tracking UAVs take into account the estimated trajectory of the target UAV while determining their own optimum paths.

The optimization problem that we should solve can be formulated as [16]:

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$$\mathbf{S}_{\text{opt}} = \arg \min_{\mathbf{S}} \|\mathbf{J}(k + N_p)\|.$$

where $\mathbf{S}$ is a matrix containing the angles that each tracking UAV needs to turn between time steps $k + 1$ to $k + N_c$, k is a time index, N_c is the control horizon, $\mathbf{J}(k)$ is the FIM for the target UAV's location/velocity, and N_p is the prediction horizon. The maximum turning angle of the tracking UAVs can be included as additional constraints on the elements of $\mathbf{S}$. In practice, the target UAV may transmit intermittent signals to minimize detectability, or only a subset of the transmitted signals by the target UAV may be detected by the tracking UAVs (e.g., due to frequency hopping of the target UAV), and localization/tracking in such scenarios is an open research area.

In Fig. 4, assuming that the signal strength observations from the target drone are continuously available at tracking drones, we compare the expected estimation errors as a function of time for receding horizon (predictive) UAV tracking with a baseline bio-inspired tracking technique (in which the tracking UAVs move toward the last estimated target location). Considering three tracking UAVs and one target UAV, the receding horizon method grants 31 percent lower error over the baseline approach by the end of simulation [16]. For reference, we have plotted the results for constrained receding horizon path planning optimization so that the rate of turn of tracking UAVs is limited to $4.5^\circ/\text{s}$, which are observed to have close to identical performance when no such constraints are introduced. We also consider two different process ($\mathbf{Q}$) and measurement ($\mathbf{R}$) noise covariance matrices for the predictive case. Comparing the performance of predictive path planning in different noise scenarios, we observe that reducing the target UAV movement process noise by half results in 47 percent lower final estimation error, while reducing the measurement noise by half results in 70 percent lower final estimation error.

INTERDICTION OF UNAUTHORIZED DRONES

After detection of unauthorized UAVs (and potentially after localizing/tracking them), there can be multiple ways to interdict unauthorized drones, depending on the vulnerability of the UAVs to different techniques. In this section, we briefly review some of the common approaches, which are also summarized in Fig. 5. One way to interdict a drone is to jam its remote control signal (commonly in the 2.4 GHz industrial, scientific, and medical, ISM, band) and/or GPS signal using a jammer gun, as illustrated in Fig. 5a. In another approach, shown in Fig. 5b, an operator fires a cannister with large nets, and as soon as the drone's rotors become tangled, a parachute brings the drone safely down to the ground. As discussed previously, certain types of drones may be vulnerable to de-authentication attacks as in Fig. 5c, which can be used for breaking the communication link between the drone and its operator.

As shown in Fig. 5d, Scotland's metropolitan police are training eagles to take down suspicious drones. This low-cost idea does not require use of any other devices and has no danger to civilians. Eagles were made to consider drones as prey so that they catch them

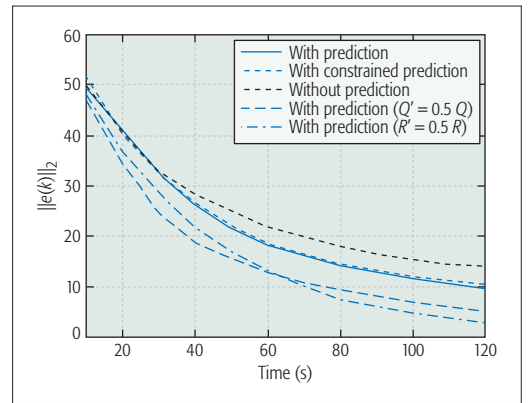


Figure 4. Tracking performance of target drone using three surveillance drones. The $\|e(k)\|_2$ in the y-axis refers to the estimation error between the true and the estimated state of the target UAV at time step k .

and place them in a safe area. One other way in which a drone could be brought down is by using another drone. For example, as shown in Fig. 5e, Tokyo police created a drone squad for privacy breaches. The police department was provided net-carrying drones that will trap suspicious drones flying in the vicinity. The GPS spoofing in Fig. 5f is another possible cyber attack that has been shown to work effectively on drones [17], which can be implemented even with low-cost SDRs. The communication links in drones include incoming signals from GPS satellites, signals indicating the drone's presence, and a two-way link between the ground station and the drone. The basic idea in GPS spoofing is to transmit fake GPS coordinates to the control system of the drone. This will hijack the drone, and subsequently it will be completely under the control of the attacker. A successful attack is conducted when the attacker is very close to the drone, or equivalently by using a directional antenna with narrow beamwidth aiming at the drone. Due to lacking an authentication mechanism, civilian drones can be attacked easily by delaying signals, whereas attacks on military drones are complicated due to the use of authentication mechanisms.

CONCLUSION

In this article, we provide a survey on cyber and physical threats that may come from small UAVs, possible ways to detect and track such small unauthorized UAVs, and common methods that are considered for interdicting such UAVs once detected. We also present our preliminary results on UWB radar-based range estimation to small UAVs, as well as receding horizon tracking of unauthorized UAVs using multiple other UAVs. Future research directions include developing effective machine learning methods for detecting, classifying, and tracking drones in various scenarios of interest, considering radar, acoustic, and RF-signal-based techniques.

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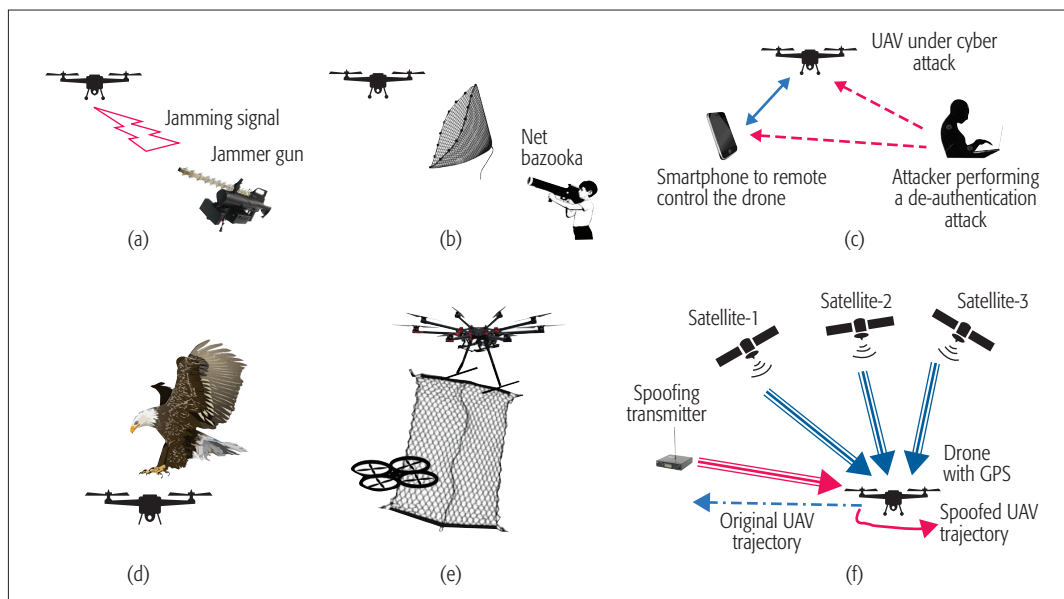


Figure 5. Different approaches for interdicting unauthorized UAVs: a) use of a drone jammer gun; b) bazooka that fires giant nets to a drone; c) de-authentication cyber attack to a drone; d) eagle trained to catch a drone and place it in a safe zone; e) net traps used by large drones to catch smaller drones; f) taking control of a drone using GPS spoofing.

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A successful attack is conducted when the attacker is very close to the drone, or equivalently by using a directional antenna with narrow beamwidth aiming at the drone. Due to no authentication mechanism, civilian drones can be attacked easily by delaying signals, whereas attacks on military drones are complicated due to use of authentication mechanisms.